MUGUNTHAN R

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mugunthan7727@gmail.com

SKILLS

Mechanical Engineering

- **CAD & Design** – SolidWorks, 3D Modeling, 2D Drafting
- **Design Practices** – DFM, DFA, Tolerance Stack-up
- **Manufacturing** – CNC Machining, Precision Turning, Assembly
- **Connector Systems** – Fiber Optic Connectors, MIL-DTL-38999
- **Enclosures** – Ruggedized Chassis, RFoF PCB Integration
- **Inspection** – CMM, Dimensional Validation, Reverse Engineering
- **Standards** – Defense & Aerospace, IP-Rated Designs

EXPERIENCE

Waveoptix Defence Solutions Pvt. Ltd. India
Mechanical Design Engineer *Sep 2023 – Present*

- Designed machined and fabricated rugged chassis for PCB-based systems
- Developed IP-rated enclosures resistant to dust, water, shock, and vibration
- Created SolidWorks 3D models, 2D drawings, and BOMs
- Performed reverse engineering using CMM for dimensional validation
- Worked on fiber optic and MIL-DTL-38999 connector designs
- Coordinated with local vendors for CNC machining, assembly, and inspection

Electrowing Technologies Pvt. Ltd. Bangalore, India
Mechanical Design Engineer Intern *Dec 2020 – Aug 2023*

- Developed 3D models and manufacturing drawings using SolidWorks
- Applied DFM and DFA principles for precision components
- Prepared BOMs and supported CNC machining and fabrication
- Coordinated with vendors and supported testing activities

PROJECTS

Waveoptix Defence Solutions Pvt. Ltd.
Ruggedized Chassis Design – RFoF PCBs *Current*

- Designed rugged CNC-machined enclosures for RF over Fiber (RFoF) PCB assemblies
- Integrated PCB mounts, connector interfaces, sealing, and grounding features
- Supported manufacturing, assembly, and inspection

Waveoptix Defence Solutions Pvt. Ltd.
Fiber Optic & MIL-DTL-38999 Connector Development *End-to-End*

- Developed a 4-channel fiber optic connector from concept to production
- Designed connector shells, alignment features, and coupling mechanisms
- Supported machining, assembly, and testing activities

EDUCATION

PA College of Engineering and Technology
Bachelor of Engineering – Mechanical Engineering *2020*

NV Polytechnic College
Diploma in Mechanical Engineering *2015*